Probability Distributions

Type I: Discrete random variable

1. The probability mass function of a random variable is 0, except at the points X=0,1,2. At these points, it has the values
$P(0) = 3C^3, P(1) = 4C - 10C^2, P(2) = 5C - 1$
(i) Determine C.
[N19/Extc/5M]
(ii) Find probability of $P(X < 1)$
[M18/Extc/5M]
(iii) $P(1 < X \leq 2), P(0 < X \leq 2)$
[N18/IT/5M]
Solution:
It is given,

X	0	1	2
$P(X = x)$	$3C^3$	$4C - 10C^2$	$5C - 1$

We know that
$\sum P(X) = 1$
$3C^3 + 4C - 10C^2 + 5C - 1 = 1$
$3C^3 - 10C^2 + 9C - 2 = 0$
$C = 2,1,\frac{1}{3}$
We see that $C = \frac{1}{3}$ satisfies the necessary condition of a p.d.

X	0	1	2
$P(X = x)$	$\frac{1}{9}$	$\frac{2}{9}$	$\frac{2}{3}$

$P(X < 1) = P(0) = \frac{1}{9}$
$P(1 < X \leq 2) = P(2) = \frac{2}{3}$
$P(0 < X \leq 2) = P(1) + P(2) = \frac{2}{9} + \frac{2}{3} = \frac{8}{9}$

2.

X	:	0	1	2	3	4	5	6	7
P(X)	:	0	c	2c	2c	3c	c^2	$2c^2$	$7c^2$+c , Find c

(i) Find $P(X \geq 6)$
(ii) Find P (X< 6)
(iii) Find k if $P(X \leq k) > \frac{1}{2}$ where k is a positive integer.

(iv) Find$P\left(\frac{1.5<X<4.5}{X>2}\right)$

(v) Find$P\left(\frac{-1\le X\le 1}{-2\le X\le 3}\right)$ Ans. 0.1, 0.19, 0.81, 4, 5/7, 1/5

3. A random variable X takes values 1, 2, 3, 4 such that $2P(X=1)=3P(X=2)=P(X=3)=5P(X=4)$. Find the probability distribution and the cumulative distribution function.

[M18/Inst/5M][N18/Biom/5M]

Solution:

Let $2P(X=1)=3P(X=2)=P(X=3)=5P(X=4)=k$

$\therefore P(X=1)=\frac{k}{2}$

$\therefore P(X=2)=\frac{k}{3}$

$\therefore P(X=3)=k$

$\therefore P(X=4)=\frac{k}{5}$

We have,

$\sum P(X)=1$

$\frac{k}{2}+\frac{k}{3}+k+\frac{k}{5}=1$

$\frac{61}{30}k=1$

$k=\frac{30}{61}$

Thus,

X	:	1	2	3	4
P(X)	:	$\frac{15}{61}$	$\frac{10}{61}$	$\frac{30}{61}$	$\frac{6}{61}$
F(X)	:	$\frac{15}{61}$	$\frac{25}{61}$	$\frac{55}{61}$	1

4. A random variable X has the following probability function

X	:	1	2	3	4	5	6	7
P(X)	:	k	2k	3k	k^2	k^2+k	$2k^2$	$4k^2$

Find (i) k (ii) $P(X<5)$ (iii) $P(X>5)$ (iv) $P\left(\frac{X<5}{2<X\le 6}\right)$ (v) $P(0<X<6)$

[N18/Inst/8M][M19/Chem/5M] [N19/IT/5M]

Solution:

We have,

$\sum P(X)=1$

$k+2k+3k+k^2+k^2+k+2k^2+4k^2=1$

$8k^2+7k-1=0$

$k = -1, \frac{1}{8}$

We see that $k = \frac{1}{8}$ satisfies the necessary condition of a pd

Thus,

X	:	1	2	3	4	5	6	7
P(X)	:	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{3}{8}$	$\frac{1}{64}$	$\frac{9}{64}$	$\frac{1}{32}$	$\frac{1}{16}$

$P(X < 5) = \frac{1}{8} + \frac{1}{4} + \frac{3}{8} + \frac{1}{64} = \frac{49}{64}$

$P(X > 5) = \frac{1}{32} + \frac{1}{16} = \frac{3}{32}$

$$P\left(\frac{X < 5}{2 < X \le 6}\right) = \frac{P[(X<5)\cap(2<X\le6)]}{P(2<X\le6)}$$
$$= \frac{P[(1,2,3,4)\cap(3,4,5,6)]}{P(3,4,5,6)}$$
$$= \frac{P(3,4)}{P(3,4,5,6)}$$
$$= \frac{\frac{3}{8}+\frac{1}{64}}{\frac{3}{8}+\frac{1}{64}+\frac{9}{64}+\frac{1}{32}}$$
$$= \frac{25}{36}$$

$P(0 < X < 6) = \frac{1}{8} + \frac{1}{4} + \frac{3}{8} + \frac{1}{64} + \frac{9}{64} = \frac{29}{32}$

5. The p.d.f of a random variable X is

X	:	0	1	2	3	4	5	6
P(X = x)	:	k	3k	5k	7k	9k	11k	13k

Find k, $P(X < 4), P(3 < X \le 6)$

[N15/AutoMechCivil/6M][M19/IT/5M]

Solution:

$\sum p(x) = 1$

$k + 3k + 5k + 7k + 9k + 11k + 13k = 1$

$49k = 1$

$\therefore k = \frac{1}{49}$

X	0	1	2	3	4	5	6
P(X)	$\frac{1}{49}$	$\frac{3}{49}$	$\frac{5}{49}$	$\frac{7}{49}$	$\frac{9}{49}$	$\frac{11}{49}$	$\frac{13}{49}$

$$P(X < 4) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3)$$
$$= \frac{1}{49} + \frac{3}{49} + \frac{5}{49} + \frac{7}{49}$$

$\therefore P(X < 4) = \frac{16}{49}$

$P(3 < X \le 6) = P(X = 4) + P(X = 5) + P(X = 6) = \frac{9}{49} + \frac{11}{49} + \frac{13}{49}$

$\therefore P(3 < X \leq 6) = \frac{33}{49}$

6. The p.d.f of a random variable X is

X	:	0	1	2	3	4
P(X = x)	:	$\frac{1}{16}$	4k	6k	4k	k

Find k, $P(X < 4), P(X > 3), P(0 < X \leq 2)$
[N19/AutoMechCivil/5M]
Solution:
We have,
$\sum P(X) = 1$
$\frac{1}{16} + 4k + 6k + 4k + k = 1$
$15k = \frac{15}{16}$
$k = \frac{1}{16}$
Thus,

X	:	0	1	2	3	4
P(X)	:	$\frac{1}{16}$	$\frac{1}{4}$	$\frac{3}{8}$	$\frac{1}{4}$	$\frac{1}{16}$

$P(X < 4) = \frac{1}{16} + \frac{1}{4} + \frac{3}{8} + \frac{1}{4} = \frac{15}{16}$
$P(X > 3) = \frac{1}{16}$
$P(0 < X \leq 2) = \frac{1}{4} + \frac{3}{8} = \frac{5}{8}$

7. A discrete random variable has the p.d.f given by

X	:	-2	-1	0	1	2	3
P(X)	:	0.2	k	0.1	2k	0.1	2k

Find k, mean & variance.
[N15/CompIT/5M][M17/CompIT/6M][M18/Comp/6M][M18/IT/5M]
[N18/AutoMechCivil/6M][N18/Comp/6M][N19/Comp/6M]
Solution:
We have,
$\sum p(x) = 1$
$0.2 + k + 0.1 + 2k + 0.1 + 2k = 1$
$0.4 + 5k = 1$
$5k = 0.6$
$\therefore k = \frac{3}{25} = 0.12$

The p.d.f becomes,

X	-2	-1	0	1	2	3
P(X)	0.2	0.12	0.1	0.24	0.1	0.24

Mean $= E(X) = \sum x \,.\, p(x)$

$$= (-2)(0.2) + (-1)(0.12) + (0)(0.1) + (1)(0.24) + (2)(0.1) + (3)(0.24)$$

$$= \frac{16}{25} = 0.64$$

$$E(X^2) = \sum x^2 . p(x)$$

$$= (-2)^2(0.2) + (-1)^2(0.12) + (0)^2(0.1) + (1)^2(0.24) + (2)^2(0.1) + (3)^2(0.24)$$

$$= \frac{93}{25} = 3.72$$

$$V(X) = E(X^2) - [E(X)]^2 = 3.72 - (0.64)^2 = 3.3104$$

8. If the mean of the following p.d is 16 find m, n and variance.

X	:	8	12	16	20	24
P(X)	:	1/8	m	n	1/4	1/12

[N16/AutoMechCivil/5M][M17/AutoMechCivil/5M][N19/Chem/4M]

Solution:

We have,

$$\sum p(x) = 1$$

$$\frac{1}{8} + m + n + \frac{1}{4} + \frac{1}{12} = 1$$

$$m + n = \frac{13}{24} \text{ ……………(1)}$$

Also,

$$E(X) = 16$$

$$\sum x . p(x) = 16$$

$$8\left(\frac{1}{8}\right) + 12m + 16n + 20\left(\frac{1}{4}\right) + 24\left(\frac{1}{12}\right) = 16$$

$$12m + 16n = 8 \text{ ……….(2)}$$

Solving equations (1) & (2), we get

$$m = \frac{1}{6}, n = \frac{3}{8}$$

Now,

$$E(X^2) = \sum x^2 p(x)$$

$$= 8^2\left(\frac{1}{8}\right) + 12^2 m + 16^2 n + 20^2\left(\frac{1}{4}\right) + 24^2\left(\frac{1}{12}\right)$$

$$= 276$$

$$V(X) = E(X^2) - [E(X)]^2 = 276 - 16^2 = 20$$

9. Let X, Y be the numbers obtained on two dice when thrown. Find the expectation of (i) X (ii) X – Y (iii) 2X – 3Y. Ans. 3.5, 0, -3.5

10. If X_1 has mean 5 and variance 5, X_2 has mean 2 and variance 3. If X_1 and X_2 are independent random variables, find
(i) $E(X_1 + X_2), V(X_1 + X_2)$
(ii) $E(2X_1 + 3X_2 - 5), V(2X_1 + 3X_2 - 5)$ Ans. 7, 8, 11, 47

11. Let X be a random variable with pdf

X	-3	6	9
P(X)	1/6	1/2	1/3

Find mean, variance and also find $E(2X^2 + 1)$ Ans. 5.5, 16.25, 94

12. A random variable X has the probability distribution $P(X = x) = \frac{1}{8}\ {}^3C_x$ $x = 0,1,2,3$. Find mean and variance
[M19/Comp/6M]
Solution:
The pdf is given as follows

X	0	1	2	3
$P(X = x)$	$\frac{^3C_0}{8} = \frac{1}{8}$	$\frac{^3C_1}{8} = \frac{3}{8}$	$\frac{^3C_2}{8} = \frac{3}{8}$	$\frac{^3C_3}{8} = \frac{1}{8}$

Now,
Mean, $E(x) = \sum x\,p(x)$
$E(x) = 0 \times \frac{1}{8} + 1 \times \frac{3}{8} + 2 \times \frac{3}{8} + 3 \times \frac{1}{8}$
$E(x) = 1.5$
$E(x^2) = \sum x^2 p(x)$
$E(x^2) = 0^2 \times \frac{1}{8} + 1^2 \times \frac{3}{8} + 2^2 \times \frac{3}{8} + 3^2 \times \frac{1}{8}$
$E(x^2) = 3$
Variance, $V(x) = E(x^2) - [E(x)]^2$
$V(x) = 3 - (1.5)^2 = 0.75$

13. The number of messages sent per hour over a computer network has the following probability distribution

x	10	11	12	13	14	15
P(X = x)	0.08	3k	6k	4k	4k	0.07

Find the mean and variance of number of messages sent per hour
[N18/Elex/5M]
Solution:
We have,
$\sum p(x) = 1$

$0.08 + 3k + 6k + 4k + 4k + 0.07 = 1$
$17k = \frac{17}{20}$
$\therefore k = \frac{1}{20}$
$E(X) = \sum x.p(x)$
$E(X) = 10(0.08) + 11\left(\frac{3}{20}\right) + 12\left(\frac{6}{20}\right) + 13\left(\frac{4}{20}\right) + 14\left(\frac{4}{20}\right) + 15(0.07)$
$E(X) = \frac{25}{2}$
Mean = 12.5
$E(X^2) = \sum x^2 p(x)$
$E(X^2) = 10^2(0.08) + 11^2\left(\frac{3}{20}\right) + 12^2\left(\frac{6}{20}\right) + 13^2\left(\frac{4}{20}\right) + 14^2\left(\frac{4}{20}\right) + 15^2(0.07)$
$= 158.1$
$V(X) = E(X^2) - [E(X)]^2 = 158.1 - 12.5^2 = 1.85$

14. If X_1 has mean 4 and variance 9, X_2 has mean -2 and variance 4. If X_1 and X_2 are independent random variables, find $E(2X_1 + X_2 - 3)$ and $V(2X_1 + X_2 - 3)$
[M19/Extc/5M]
Solution:
$E(X_1) = 4, V(X_1) = 9, E(X_2) = -2, V(X_2) = 4$
Now,
$E(2X_1 + X_2 - 3) = 2E(X_1) + E(X_2) - 3 = 2(4) + (-2) - 3 = 3$
$V(2X_1 + X_2 - 3) = 2^2 V(X_1) + V(X_2) = 4(9) + 4 = 40$

15. A random variable X has the following pmf – $P(X = x) = \frac{x}{15}$ for $x = 1,2,3,4,5$ find $E(X), E(2X^2 + 1)$ and $E(3X + 8)$ Ans. 3.67, 31, 19

Type II: Continuous Random Variable

1. A continuous random variable has the following probability function, $f(x) = kx^2, 0 \le x \le 2$. Determine k and find the probabilities that
   (i) $0.2 \le x \le 0.5$
   (ii) $x \ge 0.75$ given that $x \ge 0.5$ Ans. 3/8, 0.0146, 0.963
2. The diameter say X of an electric cable is assumed to be a continuous random variable with pdf; $f(x) = 6x(1-x); 0 \le x \le 1$.
   (i) Is it probability distribution function?
   (ii) Obtain cumulative distribution function
   (iii) Compute $P\left\{\frac{X \le \frac{1}{2}}{\frac{1}{3} \le X \le \frac{2}{3}}\right\}$
   (iv) determine k so that $P(X < k) = P(X > k)$
   Ans. $3x^2 - 2x^3, \frac{1}{2}, \frac{1}{2}$
3. Let X be a continuous random variable with pdf
   $f(x)$ = ax , $0 \le x \le 1$
   = a , $1 \le x \le 2$
   = $-ax + 3a$, $2 \le x \le 3$
   = 0 , elsewhere
   Determine the constant a and compute $P(X \le 1.5)$ Ans. $\frac{1}{2}, \frac{1}{2}$
4. If the probability density of a random variable is given by
$$f(x) = \begin{cases} kx & 0 \le x \le 2 \\ 2k & 2 \le x \le 4 \\ 6k - kx & 4 \le x \le 6 \end{cases}$$
   Find (i) k (ii) $P(1 \le x \le 3)$
   [M19/AutoMechCivil/5M]
   Solution:
   We know that,
$$\int f(x)dx = 1$$
$$\int_0^2 kx dx + \int_2^4 2k dx + \int_4^6 (6k - kx) dx = 1$$
$$k\left[\frac{x^2}{2}\right]_0^2 + 2k[x]_2^4 + k\left[6x - \frac{x^2}{2}\right]_4^6 = 1$$
$$k[2] + 2k[2] + k[2] = 1$$
$$8k = 1$$
$$k = \frac{1}{8}$$
   Now,
$$P(1 \le x \le 3) = \int_1^3 f(x)dx$$

$P(1 \le x \le 3) = \int_1^2 kx dx + \int_2^3 2k dx$

$P(1 \le x \le 3) = k\left[\frac{x^2}{2}\right]_1^2 + 2k[x]_2^3$

$P(1 \le x \le 3) = \frac{3k}{2} + 2k$

$P(1 \le x \le 3) = \frac{7k}{2}$

$P(1 \le x \le 3) = \frac{7}{16}$

5. Let X be a continuous random variable with probability distribution $P(x) = \begin{cases} \frac{x}{6} + k & if\ 0 < x \le 3 \\ 0 & else \end{cases}$

Evaluate k and find $P(1 \le X \le 2)$

[N18/Extc/5M][N19/Elex/6M]

Solution:

We have,

$\int f(x)dx = 1$

$\int_0^3 \frac{x}{6} + k\ dx = 1$

$\left[\frac{x^2}{12} + kx\right]_0^3 = 1$

$\frac{9}{12} + 3k = 1$

$3k = \frac{3}{12}$

$\therefore k = \frac{1}{12}$

Now,

$P(1 \le X \le 2) = \int_1^2 f(x)dx$

$= \int_1^2 \frac{x}{6} + \frac{1}{12} dx$

$= \left[\frac{x^2}{12} + \frac{x}{12}\right]_1^2 = \frac{1}{3}$

6. A continuous random variable X takes values between 2 and 5. Its density function is $f(x) = k(1 + x)$. Find k and $P(x < 4)$ Ans. $\frac{2}{27}, \frac{16}{27}$

7. Let X be a continuous r.v. with pdf $f(x) = kx(1 - x), 0 \le x \le 1$. Find k and determine a number ‘b’ such that $P(X \le b) = P(X \ge b)$

[M15/AutoMechCivil/5M][M18/N18/Biom/5M][M19/Biom/5M][M19/Inst/5M]

Solution:
We have,
$\int f(x)dx = 1$
$\int_0^1 k(x-x^2)dx = 1$
$k\left[\frac{x^2}{2}-\frac{x^3}{3}\right]_0^1 = 1$
$k\left[\frac{1}{6}\right] = 1$
$k = 6$
Now,
$P(X \le b) = P(X \ge b)$
$\int_0^b f(x)dx = \int_b^1 f(x)dx$
$\int_0^b 6(x-x^2)dx = \int_b^1 6(x-x^2)dx$
$6\left[\frac{x^2}{2}-\frac{x^3}{3}\right]_0^b = 6\left[\frac{x^2}{2}-\frac{x^3}{3}\right]_b^1$
$3b^2 - 2b^3 = (3-2) - (3b^2 - 2b^3)$
$6b^2 - 4b^3 - 1 = 0$
$4b^3 - 6b^2 + 1 = 0$
$b = -0.366, 1.366, 0.5$
$\therefore b = 0.5$ as it lies within the range $0 \le x \le 1$

8. Find k if the following function is a probability density function
$f(x) = \begin{cases} k(1-x^2) & 0 < x < 1 \\ 0 & otherwise \end{cases}$
Also find $P(0.1 < x < 0.2)$ and $P(x > 0.5)$
[M19/Elex/6M]
Solution:
We have,
$\int f(x)dx = 1$
$\int_0^1 k(1-x^2)dx = 1$
$k\left[x-\frac{x^3}{3}\right]_0^1 = 1$
$k\left[\frac{2}{3}\right] = 1$
$k = \frac{3}{2}$
Now,
$P(0.1 < x < 0.2) = \int_{0.1}^{0.2} \frac{3}{2}(1-x^2)dx$

$P(0.1 < x < 0.2) = \frac{3}{2}\left[x - \frac{x^3}{3}\right]_{0.1}^{0.2}$

$P(0.1 < x < 0.2) = \frac{293}{2000}$

Also,

$P(x > 0.5) = \int_{0.5}^{1} \frac{3}{2}(1 - x^2)dx$

$P(x > 0.5) = \frac{3}{2}\left[x - \frac{x^3}{3}\right]_{0.5}^{1}$

$P(x > 0.5) = \frac{5}{16}$

9. The pdf of random variable X is given by $f(x) = kx^2(2 - x), 0 \le x \le 2$. Find k, mean and variance. Ans. $\frac{3}{4}, \frac{6}{5}, \frac{4}{25}$

10. A continuous random variable X has the p.d.f defined by $f(x) = A + Bx, 0 \le x \le 1$. If the mean of distribution is 1/3, find A & B.
[M14/AutoMechCivil/5M][N19/Extc/6M]
Solution:
We have,

$\int f(x)dx = 1$

$\int_0^1 (A + Bx)dx = 1$

$\left[Ax + B\frac{x^2}{2}\right]_0^1 = 1$

$A + \frac{B}{2} = 1$

$2A + B = 2$ …………….(1)

Also,

Mean $= E(x) = \frac{1}{3}$

$E(x) = \int x\, f(x)dx$

$\int_0^1 x(A + Bx)dx = \frac{1}{3}$

$\left[A\frac{x^2}{2} + B\frac{x^3}{3}\right]_0^1 = \frac{1}{3}$

$\frac{A}{2} + \frac{B}{3} = \frac{1}{3}$

$3A + 2B = 2$ ……………..(2)

Solving equation (1) & (2), we get

$A = 2, B = -2$

11. The distribution function of a r.v. X is given by $F(x) = 1 - (1+x)e^{-x}, x \geq 0$. Find the density function, mean and variance of X. Ans.xe^{-x}, 2, 2
12. Define random variable with an example. Find k if the following is a pdf. $f(x) = kxe^{-4x^2}, 0 \leq x \leq \infty$. Also find mean. Ans.8, $\frac{\sqrt{\pi}}{4}$
13. The probability density function of a random variable x is given by $f(x) = k\, e^{-\frac{x}{6}}, 0 < x < \infty$. Find the mean and standard deviation of x.
Ans. 6, 6
14. A continuous r.v. X has the density function $f(x) = kx^2(1-x), 0 \leq x \leq 1$. Find k and $P(|x-\mu| \geq 2\sigma)$ where μ and σ^2 are mean and variance of X.
Ans. 12, $\frac{17}{625}$
15. The daily consumption of electric power (in million kwh) is a random variable X with probability distribution function

$$f(x) = \begin{cases} kxe^{-x/3} & for\ x > 0 \\ 0 & for\ x \leq 0 \end{cases}$$

Find the value of k, the expectation and the probability that on a given day the electric consumption is more than the expected value.

[M14/CompIT/6M]

Solution:

We have,

$$\int f(x)dx = 1$$

$$\int_0^\infty kx\, e^{-\frac{x}{3}} dx = 1$$

$$k\left[x\left(\frac{e^{-\frac{x}{3}}}{-\frac{1}{3}}\right) - (1)\left(\frac{e^{-\frac{x}{3}}}{\frac{1}{9}}\right)\right]_0^\infty = 1$$

$$k\left[0 - 0 - 0 + \frac{e^0}{\frac{1}{9}}\right] = 1$$

$$9k = 1$$

$$k = \frac{1}{9}$$

Expectation,

$$E(x) = \int x\, f(x) dx$$

$$E(x) = \int_0^\infty \frac{1}{9}\, x^2 e^{-\frac{x}{3}} dx$$

$$E(x) = \frac{1}{9}\left[x^2\left(\frac{e^{-\frac{x}{3}}}{-\frac{1}{3}}\right) - (2x)\left(\frac{e^{-\frac{x}{3}}}{\frac{1}{9}}\right) + (2)\left(\frac{e^{-\frac{x}{3}}}{-\frac{1}{27}}\right)\right]_0^\infty$$

$E(x) = \frac{1}{9}\left[0 - 0 + 0 - 0 + 0 - 2\left(\frac{1}{-\frac{1}{27}}\right)\right]$

$E(x) = 6$

Probability that on a given day the consumption is more than the expected value is given by,

$P\big(x > E(x)\big) = P(x > 6) = \int_6^\infty f(x)dx$

$= \int_6^\infty \frac{1}{9}\ x\, e^{-\frac{x}{3}}dx$

$= \frac{1}{9}\left[x\left(\frac{e^{-\frac{x}{3}}}{-\frac{1}{3}}\right) - (1)\left(\frac{e^{-\frac{x}{3}}}{\frac{1}{9}}\right)\right]_6^\infty$

$= \frac{1}{9}\left[0 - 0 - 6\left(\frac{e^{-2}}{-\frac{1}{3}}\right) + \left(\frac{e^{-2}}{\frac{1}{9}}\right)\right] = 3e^{-2} = 0.406$

16. A continuous random variable has p.d.f $f(x) = kx^3, 0 \le x \le 1$
Hence, find k, mean and $P(0.3 < x < 0.6)$
[N19/Elect/5M]

Solution:

We have,

$\int f(x)dx = 1$

$\int_0^1 kx^3 dx = 1$

$k\left[\frac{x^4}{4}\right]_0^1 = 1$

$k\left[\frac{1}{4}\right] = 1$

$k = 4$

Mean,

$E(x) = \int x\, f(x)dx$

$E(x) = \int_0^1 x.4x^3 dx$

$= 4\left[\frac{x^5}{5}\right]_0^1$

$E(x) = \frac{4}{5}$

$P(0.3 < x < 0.6) = \int_{0.3}^{0.6} 4x^3 dx = 4\left[\frac{x^4}{4}\right]_{0.3}^{0.6} = 0.1215$

17. A continuous random variable has p.d.f $f(x) = 6(x - x^2), 0 \le x \le 1$
Find mean and variance.

[N14/AutoMechCivil/5M][M15/CompIT/6M][N19/Chem/4M]

Solution:

Mean,

$E(x) = \int x\, f(x)dx$

$E(x) = \int_0^1 x.6(x - x^2)dx$

$= 6\left[\frac{x^3}{3} - \frac{x^4}{4}\right]_0^1$

$E(x) = \frac{1}{2}$

And,

$E(x^2) = \int x^2 f(x)dx$

$E(x^2) = \int_0^1 x^2.6(x - x^2)dx$

$= 6\left[\frac{x^4}{4} - \frac{x^5}{5}\right]_0^1$

$E(x^2) = \frac{3}{10}$

Variance,

$V(x) = E(x^2) - [E(x)]^2 = \frac{3}{10} - \frac{1}{4} = \frac{1}{20}$

18. Find k and then $E(X)$ for the pdf

$f(x) = \begin{cases} k(x - x^2) & 0 \le x \le 1, k \ge 0 \\ 0 & otherwise \end{cases}$

[N16/CompIT/5M]

Solution:

We have,

$\int f(x)dx = 1$

$\int_0^1 k(x - x^2)dx = 1$

$k\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = 1$

$k\left[\frac{1}{6}\right] = 1$

$k = 6$

Mean,

$E(x) = \int x\, f(x)dx$

$E(x) = \int_0^1 x.6(x - x^2)dx$

$= 6\left[\frac{x^3}{3} - \frac{x^4}{4}\right]_0^1$

$E(x) = \frac{1}{2}$

19. If p.d.f is given by $f(x) = \begin{cases} kx^2(1-x^3) & , \ 0 \le x \le 1 \\ 0 & , \quad elsewhere \end{cases}$

Find k , P(0 < x < 1/2), E(x), Variance.

[N17/Chem/5M]

Solution:

We have,

$\int f(x)dx = 1$

$\int_0^1 k(x^2 - x^5)dx = 1$

$k\left[\frac{x^3}{3} - \frac{x^6}{6}\right]_0^1 = 1$

$k\left[\frac{1}{6}\right] = 1$

$k = 6$

$P\left(0 < x < \frac{1}{2}\right) = \int_0^{\frac{1}{2}} 6(x^2 - x^5)dx = 6\left[\frac{x^3}{3} - \frac{x^6}{6}\right]_0^{\frac{1}{2}} = \frac{15}{64}$

Mean,

$E(x) = \int x\, f(x)dx$

$E(x) = \int_0^1 x.6(x^2 - x^5)dx$

$\quad = 6\left[\frac{x^4}{4} - \frac{x^7}{7}\right]_0^1$

$E(x) = \frac{9}{14}$

Also,

$E(x^2) = \int x^2\, f(x)dx$

$E(x^2) = \int_0^1 x^2.6(x^2 - x^5)dx$

$\quad = 6\left[\frac{x^5}{5} - \frac{x^8}{8}\right]_0^1$

$E(x) = \frac{9}{20}$

Variance, $V(x) = E(x^2) - E(x) = \frac{9}{20} - \left(\frac{9}{14}\right)^2 = \frac{9}{245}$

20. A continuous random variable has p.d.f, $f(x) = kx^2(1-x^3), 0 \le x \le 1$ and $f(x) = 0$, otherwise. Find k and mean

[N17/AutoMechCivil/5M] [M19/Elect/5M]

Solution:

We have,

$\int f(x)dx = 1$

$\int_0^1 k(x^2 - x^5)dx = 1$

$k\left[\frac{x^3}{3} - \frac{x^6}{6}\right]_0^1 = 1$

$k\left[\frac{1}{6}\right] = 1$

$k = 6$

Mean,

$E(x) = \int x\, f(x)dx$

$E(x) = \int_0^1 x.6(x^2 - x^5)dx$

$= 6\left[\frac{x^4}{4} - \frac{x^7}{7}\right]_0^1$

$E(x) = \frac{9}{14}$

21. If x is a continuous random variable with the probability density function given by $f(x) = \begin{cases} k(x - x^3) & ,\ 0 \le x \le 1 \\ 0 & ,\quad otherwise \end{cases}$

Find k and the mean

[M16/CompIT/5M]

Solution:

We have,

$\int f(x)dx = 1$

$\int_0^1 k(x - x^3)dx = 1$

$k\left[\frac{x^2}{2} - \frac{x^4}{4}\right]_0^1 = 1$

$k\left[\frac{1}{4}\right] = 1$

$k = 4$

Mean,

$E(x) = \int x\, f(x)dx$

$E(x) = \int_0^1 x.4(x - x^3)dx = 4\left[\frac{x^3}{3} - \frac{x^5}{5}\right]_0^1 = \frac{8}{15}$

22. The probability density function of a random variable X is $f(x) = kx^2(1 - x^3), 0 \le x \le 1$. Find k, expectation and variance

[M18/AutoMechCivil/5M][N18/Inst/6M]

Solution:

We have,

$\int f(x)dx = 1$

$\int_0^1 k(x^2 - x^5)dx = 1$

$k\left[\frac{x^3}{3} - \frac{x^6}{6}\right]_0^1 = 1$

$k\left[\frac{1}{6}\right] = 1$

$k = 6$

Mean,

$E(x) = \int x\, f(x)dx$

$E(x) = \int_0^1 x.\,6(x^2 - x^5)dx$

$= 6\left[\frac{x^4}{4} - \frac{x^7}{7}\right]_0^1$

$E(x) = \frac{9}{14}$

Also,

$E(x^2) = \int x^2\, f(x)dx$

$E(x^2) = \int_0^1 x^2.\,6(x^2 - x^5)dx$

$= 6\left[\frac{x^5}{5} - \frac{x^8}{8}\right]_0^1$

$E(x) = \frac{9}{20}$

Variance, $V(x) = E(x^2) - E(x) = \frac{9}{20} - \left(\frac{9}{14}\right)^2 = \frac{9}{245}$

23. A continuous r.v. X has p.d.f $f(x) = kx.\,e^{-x}, x \geq 0$
Find k, mean
[M18/Elect/5M]
Solution:
We have,

$\int f(x)dx = 1$

$\int_0^\infty kx\, e^{-x}dx = 1$

$k\int_0^\infty e^{-x}\, x^1 dx = 1$

$k\Gamma 2 = 1 \qquad \because \int_0^\infty e^{-x}x^n dx = \Gamma(n+1)$

$k(1) = 1$

$k = 1$

Mean,

$E(x) = \int x\, f(x)dx$

$E(x) = \int_0^\infty x^2 e^{-x}dx$

$E(x) = \int_0^\infty e^{-x}x^2dx$
$E(x) = \Gamma 3$
$E(x) = 2!$
$E(x) = 2$

24. A continuous r.v. X has p.d.f $f(x) = kx^2.e^{-x}, x \geq 0$
Find k, mean & variance.
[N14/CompIT/6M][N18/Elect/5M][N18/MTRX/6M]
Solution:
We have,
$\int f(x)dx = 1$
$\int_0^\infty kx^2\ e^{-x}dx = 1$
$k\int_0^\infty e^{-x}x^2dx = 1$
$k\ \Gamma(3) = 1$ since, $\int_0^\infty e^{-x}x^n dx = \Gamma(n+1)$
$k\ (2!) = 1$
$k = \frac{1}{2}$
Expectation, $E(x) = \int x\ f(x)dx$
$E(x) = \int_0^\infty \frac{1}{2}\ x^3e^{-x}dx$
$E(x) = \frac{1}{2}\int_0^\infty e^{-x}x^3dx$
$E(x) = \frac{1}{2}\Gamma(4)$
$E(x) = \frac{3!}{2} = 3$
And, $E(x^2) = \int x^2\ f(x)dx$
$E(x^2) = \int_0^\infty \frac{1}{2}\ x^4e^{-x}dx$
$E(x^2) = \frac{1}{2}\int_0^\infty e^{-x}x^4dx$
$E(x^2) = \frac{1}{2}\Gamma(5)$
$E(x^2) = \frac{4!}{2} = 12$
Variance,
$V(x) = E(x^2) - [E(x)]^2 = 12 - 9 = 3$

25. The daily consumption of electric power (in million kwh) is a random variable X with probability distribution function

$$f(x) = \begin{cases} kxe^{-x/3} & for\ x > 0 \\ 0 & for\ x\ \leq 0 \end{cases}$$

If the production is 6 million kWh, determine the probability that there is power cut (shortage) on any given day. Ans. $\frac{1}{9}, 0.406$

26. The daily consumption of electric power (in million kwh) is a random variable X with probability distribution function

$$f(x) = \begin{cases} kxe^{-x/5} & for\ x > 0 \\ 0 & for\ x \leq 0 \end{cases}$$

Find the value of k, the expectation and the probability that on a given day the electric consumption is more than the expected value.

[M16/AutoMechCivil/6M]

Solution:

We have,

$$\int f(x)dx = 1$$

$$\int_0^\infty kx\, e^{-\frac{x}{5}}dx = 1$$

$$k\left[x\left(\frac{e^{-\frac{x}{5}}}{-\frac{1}{5}}\right) - (1)\left(\frac{e^{-\frac{x}{5}}}{\frac{1}{25}}\right)\right]_0^\infty = 1$$

$$k\left[0 - 0 - 0 + \frac{e^0}{\frac{1}{25}}\right] = 1$$

$$25k = 1$$

$$k = \frac{1}{25}$$

Expectation,

$$E(x) = \int x\, f(x)dx$$

$$E(x) = \int_0^\infty \frac{1}{25}\, x^2 e^{-\frac{x}{5}}dx$$

$$E(x) = \frac{1}{25}\left[x^2\left(\frac{e^{-\frac{x}{5}}}{-\frac{1}{5}}\right) - (2x)\left(\frac{e^{-\frac{x}{5}}}{\frac{1}{25}}\right) + (2)\left(\frac{e^{-\frac{x}{5}}}{-\frac{1}{125}}\right)\right]_0^\infty$$

$$E(x) = \frac{1}{25}\left[0 - 0 + 0 - 0 + 0 - 2\left(\frac{1}{-\frac{1}{125}}\right)\right]$$

$$E(x) = 10$$

Probability that on a given day the consumption is more than the expected value is given by,

$$P\big(x > E(x)\big) = P(x > 10) = \int_{10}^\infty f(x)dx$$

$$= \int_{10}^\infty \frac{1}{25}\, x\, e^{-\frac{x}{5}}dx$$

$$= \frac{1}{25}\left[x\left(\frac{e^{-\frac{x}{5}}}{-\frac{1}{5}}\right) - (1)\left(\frac{e^{-\frac{x}{5}}}{\frac{1}{25}}\right)\right]_{10}^\infty$$

$$= \frac{1}{25}\left[0 - 0 - 10\left(\frac{e^{-2}}{-\frac{1}{5}}\right) + \left(\frac{e^{-2}}{\frac{1}{25}}\right)\right] = 3e^{-2} = 0.406$$

27. Suppose that in a certain region, the daily rainfall (in inches) is a continuous random variable X with probability density function f(x) is given by $f(x) = \begin{cases} \frac{3}{4}(2x - x^2) & ,0 \le x \le 2 \\ 0 & ,otherwise \end{cases}$

Find the probability that on a given day in this region the rain fall is (i) not more than 1 inch (ii) greater than 1.5 inches (iii) between 0.5 and 1.5 inches **[N18/MTRX/6M]**

Solution:

$$P(not\ more\ than\ 1\ inch) = P(X < 1)$$
$$= \int_0^1 f(x)dx$$
$$= \int_0^1 \frac{3}{4}(2x - x^2)dx$$
$$= \frac{3}{4}\left[\frac{2x^2}{2} - \frac{x^3}{3}\right]_0^1$$
$$= \frac{1}{2}$$

$$P(greater\ than\ 1.5\ inch) = P(X > 1.5)$$
$$= \int_{1.5}^2 f(x)dx$$
$$= \int_{1.5}^2 \frac{3}{4}(2x - x^2)dx$$
$$= \frac{3}{4}\left[\frac{2x^2}{2} - \frac{x^3}{3}\right]_{1.5}^2$$
$$= \frac{5}{32}$$

$$P(between\ 0.5\ and\ 1.5\ inch) = P(0.5 < X < 1.5)$$
$$= \int_{0.5}^{1.5} f(x)dx$$
$$= \int_{0.5}^{1.5} \frac{3}{4}(2x - x^2)dx$$
$$= \frac{3}{4}\left[\frac{2x^2}{2} - \frac{x^3}{3}\right]_{0.5}^{1.5}$$
$$= \frac{11}{16}$$

28. The length of time in minutes, a lady speaks on telephone is found to be a r.v. with p.d.f $f(x) = \begin{cases} A\, e^{-\frac{x}{5}} & ,x \ge 0 \\ 0 & ,otherwise \end{cases}$

Find A& the probability that she will speak for more than 10 min, less than 5 min. Ans. 1/5, $1/e^2$, 1-1/e

Type III: Moments

1. The first four moments of a distribution about the value 4 are -1.5, 17, -30, 108. Calculate the moments about mean.
[M14/CompIT/4M]
Solution:
Given that,
$A = 4, \mu_1' = -1.5, \mu_2' = 17, \mu_3' = -30, \mu_4' = 108$
Now, moments about mean,
$\mu_1 = 0$
$\mu_2 = \mu_2' - 2\mu_1'\mu_1' + \mu_1'^2 = (17) - 2(-1.5)^2 + (-1.5)^2 = 14.75$
$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 3\mu_1'\mu_1'^2 - \mu_1'^3$
$\mu_3 = (-30) - 3(17)(-1.5) + 2(-1.5)^3 = 39.75$
$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 4\mu_1'\mu_1'^3 + \mu_1'^4$
$\mu_4 = (108) - 4(-30)(-1.5) + 6(17)(-1.5)^2 - 3(-1.5)^4$
$\mu_4 = 142.3125$
2. The first four moments of a distribution about the value 5 are 2, 20, 40, 50. Calculate mean, variance, μ_3, μ_4 Ans. $7, 16, -64, 162$
3. A random variable has the following p.d.f

X	-2	3	1
P(X)	1/3	1/2	1/6

Find the first four raw moments and first four central moments.
Ans. 1, 6, 11, 46 & 0, 5, -5, 35
4. A random variable X has the following p.d.:

X	0	1	2	3
P(X=x)	1/6	1/3	1/3	1/6

Find (i) first four raw moments (ii) first four central moments
Ans. $\frac{3}{2}, \frac{19}{6}, \frac{15}{2}, \frac{115}{6}, 0, \frac{11}{12}, 0, \frac{83}{48}$
5. A continuous r. v. X has the probability distribution $f(x) = \frac{4}{81}x(9 - x^2)$ when $0 \le x \le 3$ and $f(x) = 0$ otherwise. Find the first four moments about the origin and the mean.
Ans. 8/5, 3, 216/35, 27/2 & 0, 11/25, -32/875, 3693/8750

Type IV: Moment generating function

1. A random variable X has probability density function $\frac{1}{2^x}$, x = 1, 2, 3, …. Find the m.g.f and hence, find the mean and variance.
[M18/Inst/6M][M18/N18/Biom/6M][N18/Extc/6M]
Solution:
$M_0(t) = \sum p\, e^{tx}$
$M_0(t) = \sum \frac{1}{2^x} . e^{tx}$
$M_0(t) = \frac{e^t}{2} + \frac{e^{2t}}{2^2} + \frac{e^{3t}}{2^3} + \frac{e^{4t}}{2^4} + \cdots ….$
$M_0(t) = \frac{e^t}{2}\left[1 + \frac{e^t}{2} + \frac{e^{2t}}{2^2} + \frac{e^{3t}}{2^3} + \frac{e^{4t}}{2^4} + \cdots …\right]$
$M_0(t) = \frac{e^t}{2}\left[1 - \frac{e^t}{2}\right]^{-1}$
$M_0(t) = \frac{e^t}{2}\left[\frac{2-e^t}{2}\right]^{-1}$
$M_0(t) = \frac{e^t}{2}\left[\frac{2}{2-e^t}\right]$
$M_0(t) = \frac{e^t}{2-e^t}$
$\mu_1' = \left[\frac{d}{dt} M_0(t)\right]_{t=0} = \left[\frac{(2-e^t)(e^t)-e^t(-e^t)}{(2-e^t)^2}\right]_{t=0} = \left[\frac{2e^t}{(2-e^t)^2}\right]_{t=0} = 2$
Mean = 2
$\mu_2' = \left[\frac{d^2}{dt^2} M_0(t)\right]_{t=0} = \left[\frac{(2-e^t)^2(2e^t)-(2e^t)\left(2(2-e^t)(-e^t)\right)}{(2-e^t)^4}\right]_{t=0} = 6$
Variance $= \mu_2' - {\mu_1'}^2 = 6 - (2)^2 = 2$

2. If X denotes the outcome when a fair die is tossed, find m.g.f. of X and hence, find the mean and variance.
[M18/Comp/4M]
Solution:
The pdf of a fair die tossed is given by,

X	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$M_0(t) = \sum p\, e^{tx}$
$M_0(t) = e^t\left(\frac{1}{6}\right) + e^{2t}\left(\frac{1}{6}\right) + e^{3t}\left(\frac{1}{6}\right) + e^{4t}\left(\frac{1}{6}\right) + e^{5t}\left(\frac{1}{6}\right) + e^{6t}\left(\frac{1}{6}\right)$
$M_0(t) = \frac{1}{6}[e^t + e^{2t} + \cdots … ….\, e^{6t}]$

$$\mu_1' = \left[\frac{d}{dt}M_0(t)\right]_{t=0} = \left[\frac{e^t}{6}+\frac{2e^{2t}}{6}+\frac{3e^{3t}}{6}+\frac{4e^{4t}}{6}+\frac{5e^{5t}}{6}+\frac{6e^{6t}}{6}\right]_{t=0} = \frac{7}{2}$$

Mean = $\frac{7}{2}$

$$\mu_2' = \left[\frac{d^2}{dt^2}M_0(t)\right]_{t=0} = \left[\frac{e^t}{6}+\frac{4e^{2t}}{6}+\frac{9e^{3t}}{6}+\frac{16e^{4t}}{6}+\frac{25e^{5t}}{6}+\frac{36e^{6t}}{6}\right]_{t=0} = \frac{91}{6}$$

Variance $= \mu_2' - {\mu_1'}^2 = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{35}{12}$

3. A random variable has the following p.d.f

X	-2	3	1
P(X)	1/3	1/2	1/6

Find the m.g.f and hence find first four central moments.

[M15/CompIT/4M][M19/Elect/6M]

Solution:

$$M_0(t) = \sum p\, e^{tx} = e^{-2t}\left(\frac{1}{3}\right) + e^{3t}\left(\frac{1}{2}\right) + e^{t}\left(\frac{1}{6}\right) = \frac{e^{-2t}}{3}+\frac{e^{3t}}{2}+\frac{e^t}{6}$$

$$\mu_1' = \left[\frac{d}{dt}M_0(t)\right]_{t=0} = \left[-\frac{2e^{-2t}}{3}+\frac{3e^{3t}}{2}+\frac{e^t}{6}\right]_{t=0} = 1$$

Mean = 1

Thus, M.g.f. about mean,

$$M_{\overline{x}}(t) = \sum p e^{t(x-\overline{x})} = \sum p\, e^{t(x-1)}$$

$$M_{\overline{x}}(t) = e^{t(-2-1)}\left(\frac{1}{3}\right) + e^{t(3-1)}\left(\frac{1}{2}\right) + e^{t(1-1)}\left(\frac{1}{6}\right)$$

$$M_{\overline{x}}(t) = \frac{e^{-3t}}{3}+\frac{e^{2t}}{2}+\frac{1}{6}$$

Thus,

$$\mu_1 = \left[\frac{d}{dt}M_{\overline{x}}(t)\right]_{t=0} = \left[-\frac{3e^{-3t}}{3}+\frac{2e^{2t}}{2}\right]_{t=0} = 0$$

$$\mu_2 = \left[\frac{d^2}{dt^2}M_{\overline{x}}(t)\right]_{t=0} = \left[\frac{9e^{-3t}}{3}+\frac{4e^{2t}}{2}\right]_{t=0} = 5$$

$$\mu_3 = \left[\frac{d^3}{dt^3}M_{\overline{x}}(t)\right]_{t=0} = \left[-\frac{27e^{-3t}}{3}+\frac{8e^{2t}}{2}\right]_{t=0} = -5$$

$$\mu_4 = \left[\frac{d^4}{dt^4}M_{\overline{x}}(t)\right]_{t=0} = \left[\frac{81e^{-3t}}{3}+\frac{16e^{2t}}{2}\right]_{t=0} = 35$$

4. A random variable X has the following density function

$f(x) = \begin{cases} ke^{-kx}, & x > 0 \\ 0\,, & x \le 0 \end{cases}$. find mgf and hence its mean and variance

[N18/Inst/6M]

Solution:

$$M_0(t) = \int f(x) e^{tx} dx$$

$M_0(t) = \int_0^{\infty} ke^{-kx}e^{tx}dx$
$M_0(t) = k\int_0^{\infty} e^{-(k-t)x}dx$
$M_0(t) = k\left[\frac{e^{-(k-t)x}}{-(k-t)}\right]_0^{\infty}$
$M_0(t) = k\left[0 - \frac{1}{-(k-t)}\right]$
$M_0(t) = \frac{k}{k-t}$
$\mu_1' = \left[\frac{d}{dt}M_0(t)\right]_{t=0} = \left[k\left(-\frac{1}{(k-t)^2}\right)(-1)\right]_{t=0}$
$\mu_1' = \left[\frac{k}{(k-t)^2}\right]_{t=0} = \frac{1}{k}$
Mean = $\frac{1}{k}$
$\mu_2' = \left[\frac{d^2}{dt^2}M_0(t)\right]_{t=0} = \left[k\left(-\frac{2}{(k-t)^3}\right)(-1)\right]_{t=0}$
$\mu_2' = \left[\frac{2k}{(k-t)^3}\right]_{t=0} = \frac{2}{k^2}$
Variance $= \mu_2' - {\mu_1'}^2 = \frac{2}{k^2} - \left(\frac{1}{k}\right)^2 = \frac{1}{k^2}$

5. Find the moment generating function of the random variable having the following probability density function. Also find the mean and variance – $f(x) = \frac{1}{2}$, $-1 \le x \le 1$and $f(x) = 0$, elsewhere. Ans.$\frac{sinht}{t}$, 0, 1/3
6. The random variable X can assume the values 1 and -1 with probability $\frac{1}{2}$ each. Find the moment generating function and the first four moments about the origin. Ans.$cosht$, 0,1,0,1
7. A random variable X has the p.d. $P(X = x) = \frac{1}{8}\ {}^3C_x$, $x = 0, 1, 2, 3$. Find the mgf of X
[N19/Comp/4M]
Solution:

X	:	0	1	2	3
P(X)	:	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

$M_0(t) = \sum p\, e^{tx}$
$M_0(t) = e^{0t}\left(\frac{1}{8}\right) + e^{t}\left(\frac{3}{8}\right) + e^{2t}\left(\frac{3}{8}\right) + e^{3t}\left(\frac{1}{8}\right)$
$M_0(t) = \frac{1+3e^t+3e^{2t}+e^{3t}}{8}$

8. A random variable X has p.d.f $f(x) = kx^2.e^{-x}, x \geq 0$ and $f(x) = 0$ otherwise.
Find (i) k (ii) mean (iii) variance (iv) mgf (v) cdf of x (vi) $P(0 < x < 1)$
[M18/Elex/6M]
Solution:
We have,
$\int f(x)dx = 1$
$\int_0^\infty kx^2\, e^{-x}dx = 1$
$k\int_0^\infty e^{-x}x^2dx = 1$
$k\,\Gamma(3) = 1$ since, $\int_0^\infty e^{-x}x^n dx = \Gamma(n+1)$
$k\,(2!) = 1$
$k = \frac{1}{2}$
Mean, $E(x) = \int x\, f(x)dx$
$E(x) = \int_0^\infty \frac{1}{2}\, x^3 e^{-x}dx$
$E(x) = \frac{1}{2}\int_0^\infty e^{-x}x^3dx$
$E(x) = \frac{1}{2}\Gamma(4)$
$E(x) = \frac{3!}{2} = 3$
And, $E(x^2) = \int x^2\, f(x)dx$
$E(x^2) = \int_0^\infty \frac{1}{2}\, x^4 e^{-x}dx$
$E(x^2) = \frac{1}{2}\int_0^\infty e^{-x}x^4dx$
$E(x^2) = \frac{1}{2}\Gamma(5)$
$E(x^2) = \frac{4!}{2} = 12$
Variance, $V(x) = E(x^2) - [E(x)]^2 = 12 - 9 = 3$
Now, m.g.f
$M_0(t) = \int f(x)e^{tx}dx$
$M_0(t) = \int_0^\infty \frac{1}{2}x^2e^{-x}e^{tx}dx$
$M_0(t) = \frac{1}{2}\int_0^\infty x^2 e^{-(1-t)x}dx$
Put $(1-t)x = y$
$x = \frac{y}{1-t}$
$dx = \frac{dy}{1-t}$
$M_0(t) = \frac{1}{2}\int_0^\infty \frac{y^2}{(1-t)^2}.e^{-y}.\frac{dy}{1-t}$

$M_0(t) = \frac{1}{2(1-t)^3}\int_0^\infty e^{-y}y^2dy$

$M_0(t) = \frac{1}{2(1-t)^3}.\Gamma 3$ since, $\int_0^\infty e^{-x}x^n dx = \Gamma(n+1)$

$M_0(t) = \frac{1}{2(1-t)^3}.2!$

$M_0(t) = \frac{1}{(1-t)^3}$

Now, cdf

$F(x) = \int_0^x f(x)dx$

$F(x) = \int_0^x \frac{1}{2}\, x^2 e^{-x}dx$

$F(x) = \frac{1}{2}\left[x^2\left(\frac{e^{-x}}{-1}\right) - (2x)\left(\frac{e^{-x}}{1}\right) + (2)\left(\frac{e^{-x}}{-1}\right)\right]_0^x$

$F(x) = \frac{1}{2}[-x^2e^{-x} - 2xe^{-x} - 2e^{-x} + 2e^0]$

$F(x) = \frac{1}{2}[-x^2e^{-x} - 2xe^{-x} - 2e^{-x} + 2]$

Also,

$P(0 < x < 1) = \int_0^1 f(x)dx$

$= \int_0^1 \frac{1}{2}\, x^2e^{-x}dx$

$= \frac{1}{2}\left[x^2\left(\frac{e^{-x}}{-1}\right) - (2x)\left(\frac{e^{-x}}{1}\right) + (2)\left(\frac{e^{-x}}{-1}\right)\right]_0^1$

$= \frac{1}{2}[-e^{-1} - 2e^{-1} - 2e^{-1} + 2]$

$= \frac{1}{2}[2 - 5e^{-1}]$

$= 0.0803$

9. If X denotes the outcome when a fair die is tossed, find m.g.f. of X about the origin and hence, find the first two moments about the origin

[N18/Elex/4M]

Solution:

The pdf of a fair die tossed is given by,

X	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$M_0(t) = \sum p\, e^{tx}$

$M_0(t) = e^t\left(\frac{1}{6}\right) + e^{2t}\left(\frac{1}{6}\right) + e^{3t}\left(\frac{1}{6}\right) + e^{4t}\left(\frac{1}{6}\right) + e^{5t}\left(\frac{1}{6}\right) + e^{6t}\left(\frac{1}{6}\right)$

$M_0(t) = \frac{1}{6}[e^t + e^{2t} + \cdots \ldots \ldots . e^{6t}]$

$\mu_1' = \left[\frac{d}{dt}M_0(t)\right]_{t=0} = \left[\frac{e^t}{6} + \frac{2e^{2t}}{6} + \frac{3e^{3t}}{6} + \frac{4e^{4t}}{6} + \frac{5e^{5t}}{6} + \frac{6e^{6t}}{6}\right]_{t=0} = \frac{7}{2}$

$$\mu_2' = \left[\frac{d^2}{dt^2} M_0(t)\right]_{t=0} = \left[\frac{e^t}{6} + \frac{4e^{2t}}{6} + \frac{9e^{3t}}{6} + \frac{16e^{4t}}{6} + \frac{25e^{5t}}{6} + \frac{36e^{6t}}{6}\right]_{t=0} = \frac{91}{6}$$

10. A random variable has the following p.d.f

X	-2	3	1
P(X)	1/3	1/2	1/6

Find the m.g.f and hence find first four moments about the origin
[N19/Extc/8M]
Solution:

$$M_0(t) = \sum p\, e^{tx} = e^{-2t}\left(\frac{1}{3}\right) + e^{3t}\left(\frac{1}{2}\right) + e^{t}\left(\frac{1}{6}\right) = \frac{e^{-2t}}{3} + \frac{e^{3t}}{2} + \frac{e^t}{6}$$

$$\mu_1' = \left[\frac{d}{dt} M_0(t)\right]_{t=0} = \left[-\frac{2e^{-2t}}{3} + \frac{3e^{3t}}{2} + \frac{e^t}{6}\right]_{t=0} = 1$$

$$\mu_2' = \left[\frac{d^2}{dt^2} M_0(t)\right]_{t=0} = \left[\frac{4e^{-2t}}{3} + \frac{9e^{3t}}{2} + \frac{e^t}{6}\right]_{t=0} = 6$$

$$\mu_3' = \left[\frac{d^3}{dt^3} M_0(t)\right]_{t=0} = \left[\frac{-8e^{-2t}}{3} + \frac{27e^{3t}}{2} + \frac{e^t}{6}\right]_{t=0} = 11$$

$$\mu_4' = \left[\frac{d^4}{dt^4} M_0(t)\right]_{t=0} = \left[\frac{16e^{-2t}}{3} + \frac{81e^{3t}}{2} + \frac{e^t}{6}\right]_{t=0} = 46$$

11. A random variable X has the following p.d.:

X	0	1	2	3
P(X=x)	1/6	1/3	1/3	1/6

Find mgf about the origin and hence find first four raw moments
[M17/CompIT/4M]
Solution:

$$M_0(t) = \sum p\, e^{tx} = e^{0t}\left(\frac{1}{6}\right) + e^{t}\left(\frac{1}{3}\right) + e^{2t}\left(\frac{1}{3}\right) + e^{3t}\left(\frac{1}{6}\right) = \frac{1}{6} + \frac{e^t}{3} + \frac{e^{2t}}{3} + \frac{e^{3t}}{6}$$

$$\mu_1' = \left[\frac{d}{dt} M_0(t)\right]_{t=0} = \left[\frac{e^t}{3} + \frac{2e^{2t}}{3} + \frac{3e^{3t}}{6}\right]_{t=0} = \frac{3}{2}$$

$$\mu_2' = \left[\frac{d^2}{dt^2} M_0(t)\right]_{t=0} = \left[\frac{e^t}{3} + \frac{4e^{2t}}{3} + \frac{9e^{3t}}{6}\right]_{t=0} = \frac{19}{6}$$

$$\mu_3' = \left[\frac{d^3}{dt^3} M_0(t)\right]_{t=0} = \left[\frac{e^t}{3} + \frac{8e^{2t}}{3} + \frac{27e^{3t}}{6}\right]_{t=0} = \frac{15}{2}$$

$$\mu_4' = \left[\frac{d^4}{dt^4} M_0(t)\right]_{t=0} = \left[\frac{e^t}{3} + \frac{16e^{2t}}{3} + \frac{81e^{3t}}{6}\right]_{t=0} = \frac{115}{6}$$

12. A random variable X has the following p.d.:

X	0	1	2	3
P(X=x)	1/6	1/3	1/3	1/6

Find mgf about the origin and hence find first two raw moments and hence the variance

[M18/Elect/6M]

Solution:

It is given that

X	0	1	2	3
P(X=x)	1/6	1/3	1/3	1/6

$M_0(t) = \sum p\, e^{tx}$

$M_0(t) = e^{0t}\left(\frac{1}{6}\right) + e^{t}\left(\frac{1}{3}\right) + e^{2t}\left(\frac{1}{3}\right) + e^{3t}\left(\frac{1}{6}\right)$

$M_0(t) = \frac{1}{6} + \frac{e^t}{3} + \frac{e^{2t}}{3} + \frac{e^{3t}}{6}$

$\mu_1' = \left[\frac{d}{dt}M_0(t)\right]_{t=0} = \left[0 + \frac{e^t}{3} + \frac{2e^{2t}}{3} + \frac{3e^{3t}}{6}\right]_{t=0} = \frac{3}{2}$

Mean = $\frac{3}{2}$

$\mu_2' = \left[\frac{d^2}{dt^2}M_0(t)\right]_{t=0} = \left[\frac{e^t}{3} + \frac{4e^{2t}}{3} + \frac{9e^{3t}}{6}\right]_{t=0} = \frac{19}{6}$

Variance $= \mu_2' - {\mu_1'}^2 = \frac{19}{6} - \left(\frac{3}{2}\right)^2 = \frac{11}{12}$

13. A random variable X has the following density function

$f(x) = \begin{cases} 2e^{-2x}, & x > 0 \\ 0, & x \le 0 \end{cases}$. find mgf and hence its mean and variance

[M18/Extc/6M]

Solution:

$M_0(t) = \int f(x)e^{tx}dx$

$M_0(t) = \int_0^\infty 2e^{-2x}e^{tx}dx$

$M_0(t) = 2\int_0^\infty e^{-(2-t)x}dx$

$M_0(t) = 2\left[\frac{e^{-(2-t)x}}{-(2-t)}\right]_0^\infty$

$M_0(t) = 2\left[0 - \frac{1}{-(2-t)}\right]$

$M_0(t) = \frac{2}{2-t}$

$\mu_1' = \left[\frac{d}{dt}M_0(t)\right]_{t=0} = \left[2\left(-\frac{1}{(2-t)^2}\right)(-1)\right]_{t=0}$

$\mu_1' = \left[\frac{2}{(2-t)^2}\right]_{t=0} = \frac{1}{2}$

Mean = $\frac{1}{2}$

$\mu_2' = \left[\frac{d^2}{dt^2} M_0(t)\right]_{t=0} = \left[2\left(-\frac{2}{(2-t)^3}\right)(-1)\right]_{t=0}$

$\mu_2' = \left[\frac{4}{(2-t)^3}\right]_{t=0} = \frac{1}{2}$

Variance $= \mu_2' - {\mu_1'}^2 = \frac{1}{2} - \left(\frac{1}{2}\right)^2 = \frac{1}{4}$

Theory

1. A box contains 2^n tickets among which ^nC_i tickets bear the number $i = 0, 1, 2, 3, \ldots\ldots.., n$. A group of m tickets are drawn. What is the expectation of their numbers?
[M18/Inst/6M]
Solution:
Let X be a random variable which can take values 0,1,2,….n with probabilities $\frac{^nC_0}{2^n}, \frac{^nC_1}{2^n}, \frac{^nC_2}{2^n}, \ldots..$

X	0	1	2	3	…..	n
P(X=x)	$\frac{^nC_0}{2^n}$	$\frac{^nC_1}{2^n}$	$\frac{^nC_2}{2^n}$	$\frac{^nC_3}{2^n}$	…..	$\frac{^nC_n}{2^n}$

$E(x) = \sum x.p(x)$
$E(x) = \frac{1}{2^n}[0.\ ^nC_0 + 1.\ ^nC_1 + 2.\ ^nC_2 + \cdots \ldots.. + n.\ ^nC_n]$
$E(x) = \frac{1}{2^n}\left[1.n + 2.\frac{n(n-1)}{2} + 3.\frac{n(n-1)(n-2)}{3!} + \cdots \ldots. + n.1\right]$
$E(x) = \frac{n}{2^n}\left[1 + (n-1) + \frac{(n-1)(n-2)}{2!} + \cdots \ldots + 1\right]$
$E(x) = \frac{n}{2^n}[1+1]^{n-1} = \frac{n}{2^n}.2^{n-1} = \frac{n}{2}$
$\therefore E(S)$ =Expectation of sum of m tickets drawn $= m.\frac{n}{2} = \frac{mn}{2}$

2. A woman with m keys with her, wants to open the door of her house by trying the keys independently and randomly one by one. Find the mean and the variance of the no. of trials required to open the door if unsuccessful keys are kept aside.
[M16/AutoMechCivil/6M]
Solution:
Let X denote the no. of keys and the probability for any key being selected out of the total m keys is 1/m

X	1	2	3	4	…..	m
P(X=x)	1/m	1/m	1/m	1/m	…..	1/m

$E(X) = \sum p.x = \frac{1}{m}.1 + \frac{1}{m}.2 + \frac{1}{m}.3 + \cdots \ldots \ldots. + \frac{1}{m}.m$
$E(X) = \frac{1}{m}(1 + 2 + 3 + \cdots.. + m)$
$E(X) = \frac{1}{m}.\frac{m(m+1)}{2} = \frac{m+1}{2}$
$\therefore$mean is $\frac{m+1}{2}$

$E(X^2) = \sum px^2 = \frac{1}{m}.1^2 + \frac{1}{m}.2^2 + \frac{1}{m}.3^2 + \cdots \ldots .. + \frac{1}{m}.m^2$

$E(X^2) = \frac{1}{m}(1^2 + 2^2 + 3^2 + \cdots \ldots . + m^2)$

$E(X^2) = \frac{1}{m}.\left(\frac{m(m+1)(2m+1)}{6}\right) = \frac{2m^2+3m+1}{6}$

$V(X) = E(X^2) - [E(X)]^2$

$V(X) = \frac{2m^2+3m+1}{6} - \left[\frac{m+1}{2}\right]^2$

$V(X) = \frac{2m^2+3m+1}{6} - \frac{m^2+2m+1}{4}$

$V(X) = \frac{8m^2+12m+4-6m^2-12m-6}{24}$

$V(X) = \frac{2m^2-2}{24}$

$V(X) = \frac{m^2-1}{12}$

∴variance is $\frac{m^2-1}{12}$

3. Find the expectation of (i) the sum (ii) the product of the number of points on the throw of n dice.
[N18/Elect/6M]
Solution:
Let X denote the number of points on any die.

$E(X) = \sum p\,x = \frac{1}{6}.1 + \frac{1}{6}.2 + \frac{1}{6}.3 + \frac{1}{6}.4 + \frac{1}{6}.5 + \frac{1}{6}.6$

$E(X) = \frac{7}{2}$

Thus, expectation of the sum of n dice is given by,

$E(S) = n.\frac{7}{2} = \frac{7n}{2}$

Thus, expectation of product of n dice is given by,

$E(P) = \frac{7}{2}.\frac{7}{2}.\frac{7}{2} \ldots \ldots .. n\ times = \left(\frac{7}{2}\right)^n$

4. There are 10 counters in a bag, 6 of which are 5 rupees each while the remaining 4 are equal, but unknown value. If the expectation of drawing a single counter at random is 4 rupees, find the unknown value.
[N15/AutoMechCivil/5M]
Solution:
Let the unknown counter be of value y rupees.
Total Counters = 6 of 5 rupees + 4 of y rupees = 10
Let X denote the value on the counters

X	5	y
P(X)	$\frac{6}{10}$	$\frac{4}{10}$

Now,

$E(X) = 4$ (given)

$E(X) = \sum x\ p(x)$

$5\left(\frac{6}{10}\right) + y\left(\frac{4}{10}\right) = 4$

$30 + 4y = 40$

$4y = 10$

$y = 2.5$

5. Find the expectation of number of failures preceding the first success in an infinite series of independent trials with constant probabilities p & q of success and failure respectively

[N17/CompIT/6M]

Solution:

We have the following probability distribution

X	:	0	1	2	3	….
P(X=x)	:	p	qp	q^2p	q^3p	….

Since, we may get success in the first trial where the number of failures X=0 and the probability is p, we may get success in the second trial when the number of failures X=1 and the probability is qp and so on.

$$
\begin{aligned}
E(x) &= \sum p_i x_i \\
&= p(0) + qp(1) + q^2p(2) + q^3p(3) + \cdots \ldots \\
&= qp[1 + 2q + 3q^2 + \cdots.] \\
&= qp[1-q]^{-2} \\
&= qp[p]^{-2} \\
&= \frac{q}{p}
\end{aligned}
$$